

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Pharmacy SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Pharmacy Branch : Pharmacy

Semester : SEMESTER - 8

Subject Code & Name: ET - 1 - BP811ET - ADVANCED INSTRUMENTATION TECHNIQUES

Time : 3 Hours]

[Total Marks : 75

Instructions to the Students:

- 1.All questions are compulsory.
- 2.Draw diagrams / figures wherever necessary
- 3.Figures to right indicate full marks

Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

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|---|---|
| A) What is Crystallography? | 2 |
| B) Define accuracy as per ICH guidelines. | 2 |
| C) Mention any two parameters used in HPLC system suitability testing. | 2 |
| D) Define coupling constant (J value). | 2 |
| E) State importance of TMS in NMR spectroscopy. | 2 |
| F) Write one application of X-ray crystallography. | 2 |
| G) State any two parameters calibrated in a UV-Visible spectrophotometer. | 2 |
| H) What are hyphenated techniques? | 2 |
| I) Define radio-labelled antigen. | 2 |
| J) What is spin-lattice relaxation? | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

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|---|----|
| A) Explain principle of ¹ H-NMR spectroscopy. Describe chemical shift and factors affecting chemical shift with suitable examples. | 10 |
| B) Explain principle, instrumentation and applications of Thermogravimetric Analysis (TGA). | 10 |
| C) Explain principle and theory of mass spectrometry. Describe instrumentation with a neat labeled diagram. | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

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| A) Elaborate on MALDI and FAB ionization techniques. | 5 |
| B) Explain Bragg's law and its significance in X-ray diffraction. | 5 |
| C) Write a note on analytical method validation as per ICH guidelines. | 5 |
| D) Explain principle and applications of GC-MS/MS. | 5 |
| E) Describe Radio immuno assay. | 5 |
| F) Explain spin-spin decoupling in NMR. | 5 |
| G) Differentiate between DTA and DSC. | 5 |
| H) Write a note on powder X-ray diffraction method. | 5 |
| I) Explain solid phase extraction (SPE). | 5 |

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